# Galaxy Genome Annotation (GGA) environment in the cloud

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https://github.com/abims-sbr/GGA_Cloud - **support.abims@sb-roscoff.fr**

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## Context

The EOSC-Life project aims to create an open collaborative digital space for life science in the European Open Science Cloud (EOSC). The ABiMS bioinformatics platform, member of the European Marine Biological Resource Centre (EMBRC) research infrastructure, is involved in the work package 2 (WP2) dedicated to make computational tools, workflows and registries findable, accessible, interoperable and reusable (FAIR). As part of the EOSC-Life WP2, we planned to provide the GGA environment available in the cloud.

## The Galaxy Genome Annotation project

An initiative from the Galaxy and GMOD communities.

Aim of the project :
Deliver a comprehensive, scalable and easy to use genome annotation experience.

https://galaxy-genome-annotation.github.io

GGA environment offers :

- Galaxy tools and workflows
- A set of dockerized GMOD tools (Apollo, Chado, JBrowse, Tripal)
- Python libraries to make Galaxy and GMOD tools work together.

Benefits

- Automatic deployment : Human intervention reduced to the minimum
- Reproducibility: 1 genome = docker-compose.yml + raw data (fasta, gff, bam) + Galaxy workflow
- Flexibility : Customizable set of applications for each genome, all configured in a single file
- Scalability: Possibility to dispatch Docker containers on a cluster of Docker hosts (e.g. docker-swarm)

Implementation in the cloud

1 genome = 1 virtual machine (VM) = 1 set of reusable dockerized web applications
Galaxy as data loading orchestrator for administrators: tools and workflows to load data into GMOD tools.

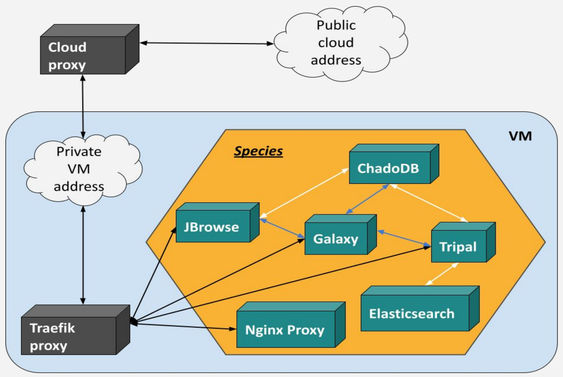



*Figure : **Schematic representation of a set of Docker containers deployed with GGA in a cloud.** Cuboids represent Docker containers. Hexagon represent a set of Docker container for a species. Black arrows represent HTTP traffic. Blue arrows represent data exchange performed using Galaxy tools. White arrows represent data exchange inherent in applications.*

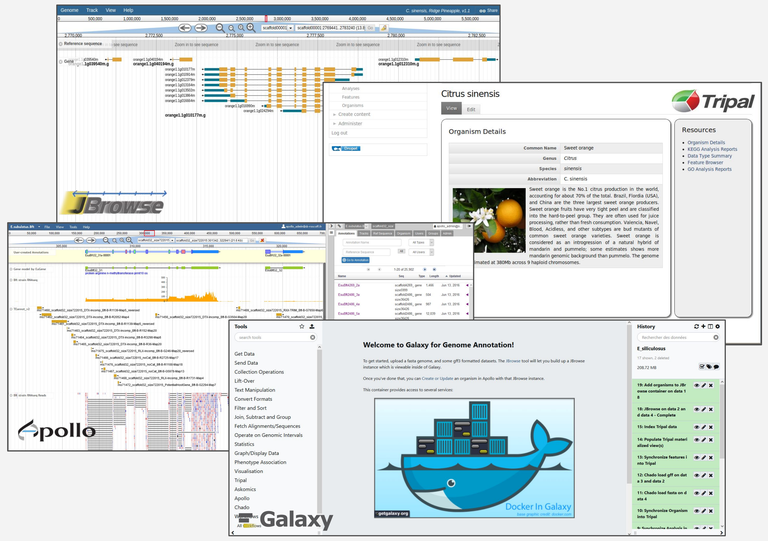
## GMOD visualisation tools deployed with GGA

GMOD makes available a collection of interconnected open-source applications and databases for visualizing, annotating, and managing genomic data.

The GMOD project provide a set of web applications for genomes :

- ***JBrowse***: genome browser
- ***Apollo***: manual curation of genome annotation
- ***Chado***: database
- ***Tripal***: gene reports with functional annotation
- ***Galaxy***: data loading orchestrator
- ***Search engine***: indexing with Elasticsearch

http://gmod.org/




## Cloud infrastructure and automation tools

Genostack is a cloud computing platform based on Openstack. It allows users to create and use virtual machine for computation.

Ansible is used for automation and Terraform for configuring and launching the VM.

***Ansible*** :
Infrastructure automation tool.
Configure systems, deploy software and orchestrate infrastructure tasks.
Works with playbooks (recipes) describing all the tasks to be performed.

***Terraform*** :
Software for building, changing, managing and versioning infrastructures.
Supports a lot of cloud infrastructure providers.

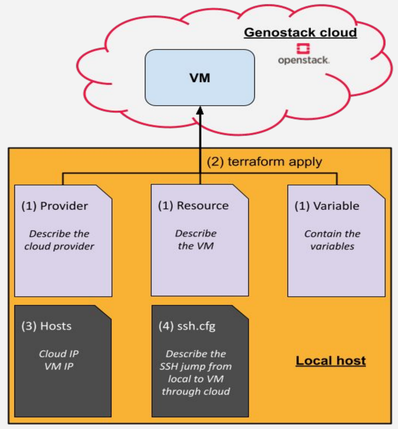
## Deployment of the GGA environment in the cloud

Ansible recipes have been developed (https://github.com/abims-sbr/GGA_Cloud) to deploy the GGA environment in an Openstack cloud infrastructure.

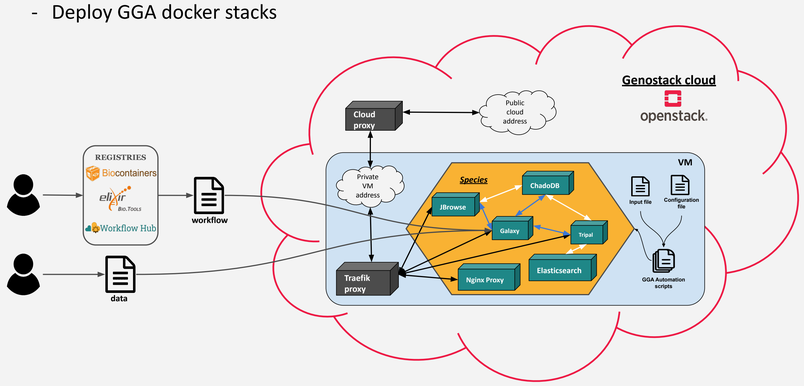
The first ansible playbook allow the deployment of a virtual machine in a cloud via the Terraform software.

What the ansible playbook do :

1. Create terraform configuration files (provider, resource and variable)
2. Apply terraform configurations to deploy the VM
3. Create ansible hosts file
4. Create ssh configuration file




A second Ansible playbook is used to automatise the deployment of the GGA environment in the virtual machine. This playbook runs the scripts presented in poster #120 and installs the dependencies required for its execution. And finally allows the user to import his data into the virtual machine to perform his genome annotation.

The ansible playbook can be separated into 4 steps :

1. Install the requirements into the VM :
   - Python 3
   - Docker
   - Git
2. Download scripts and libraries :
   - Download GGA automation scripts for deployment and data loading. (see poster #120)
   - Install python libraries
3. Deploy GGA:
   - Generate input species and configuration files
   - Deploy GGA docker stacks
4. Load data in Galaxy library from registries or user data